

Structure Factors

- ✓ It is a way used by the experimentalists to see the ordering in the system.
- ✓ $S(q)$: Measures the intensity of the scattered waves (i.e., this qty; $S(q)$ measures the intensity of the signal @ different wavelengths)

Mathematically:
$$S(q) = 1 + \frac{4\pi f}{q} \int_0^{\infty} r^2 (g(r) - 1) \frac{\sin(qr)}{qr} dr; q = \frac{2\pi}{\lambda}$$

- Setup:
- We send waves @ the system and each atom scatters them.
 - The scattered waves can constructively interfere or destructively.
 - $S(q)$: This tells us whether the interference was constructive or destructive.

$S(q) \approx 1$: No net interference

$S(q) > 1$: Constructive interference

$S(q) < 1$: Destructive interference.

Case 1: Constructive Interference; i.e. $S(q) > 1$
← one of the wavelengths

→ If atoms are separated by $d = \lambda$, the scattered waves will interfere constructively ($\because \Delta\phi = qd = \frac{2\pi}{\lambda} \times \lambda = 2\pi$)

→ This is like: Seeing a peak in $g(r)$ @ $r = d$. Now, the $d = \lambda$ will also have show a corresponding peak in $S(q)$ (but peaks can sometimes & the $S(q)$ as well)

Case 2: Destructive Interference; i.e. $S(q) < 1$

We can look @ it for 2 cases (since we're $g(r) \neq 1$ & $\sin(qr)$ terms)

(i) low q : for low q values, $g(r) - 1 < 0$ (due to excluded volume) is enough to guarantee that $S(q) < 0$ ($\because \sin(qr) > 0$)

(ii) Intermediate q : The wavelength is such that it doesn't match the interparticle distances.

Case 3: $S(\epsilon) = 1$ (No net interference)

- This happens in the random parking limit; some interfere constructively; some interfere destructively; so the resultant is no net interference.